

Final Report

Superstore Sales Performance Analysis

1. Project Name

Superstore Sales Performance Analysis

2. Project Description

This project aims to analyze sales data from a US-based Superstore between 2014 and 2017 to uncover trends in performance across sales, profit, shipping, product categories, and regions. The goal is to generate actionable insights and strategic recommendations to support business decision-making.

3. Project Objectives

- Analyze multi-year sales and profitability trends.
- Evaluate product and category-level performance.
- Investigate regional and shipping-based performance variations.
- Visualize insights using dashboards and reports.
- Recommend actions to enhance business outcomes.

- Tools Used:

-  **SQL** – data cleaning & preprocessing
-  **Python (Pandas)** – data cleaning & preprocessing
-  **Tableau** – data visualization
-  **Excel** – initial summary statistics
-  **Figma** – dashboard UI design
-  **Draw.io** – dashboard sketching
-  **Notion** – workflow documentation

Data Overview

- **Source:** Superstore dataset (Kaggle)
- **Records:** 9,994 rows



Column Name	Description
Row ID	Unique identifier for each row.
Order ID	Unique identifier for each order placed by a customer.
Order Date	Date when the order was placed.
Ship Date	Date when the product was shipped.
Ship Mode	Shipping method chosen by the customer (e.g., Standard, Express).
Customer ID	Unique identifier for each customer.
Customer Name	Name of the customer.
Segment	Customer segment (e.g., Consumer, Corporate, Home Office).
Country	Country where the customer resides.
City	City where the customer resides.
State	State where the customer resides.
Postal Code	Postal code of the customer's location.
Region	Geographical region of the customer.
Product ID	Unique identifier for each product.
Category	Category of the product (e.g., Furniture, Office Supplies).
Sub-Category	Sub-category within the product category.
Product Name	Name of the product.
Sales	Revenue generated from the sale of the product.
Quantity	Number of units sold.
Discount	Discount applied to the product.
Profit	Profit or loss incurred from the sale.

5. Project Workflow

◊ Data Collection

- Imported and explored the Superstore dataset (2014–2017).



◇ Data Cleaning

- Handled missing values, duplicates, and inconsistent types.
- Standardized date formats and categorical values using Python.

◇ Data Modeling

- Built a **Star Schema** with Fact and Dimension tables for analysis.

◇ Exploratory Data Analysis (EDA)

- Investigated trends in time, regions, shipping methods, categories, and customer behavior.
- Identified seasonal, regional, and categorical performance differences.

◇ Visualization & Reporting

- Developed interactive dashboards in Tableau.
- Designed layout in Figma and documented process in Notion.

6. Key Findings

📅 Time-Based Insights

- **Best Year:** 2017 (highest in profit, sales, orders, and quantity).
- **Seasonality:**
 - Peak months: **November–December**
 - Weakest month: **February**
- **Profit Trends:**
 - Highest monthly profit: March 2017
 - Lowest: January 2015 (−\$3,281)

🛒 Product & Category Performance

Metric	Top Category	Top Sub-Category	Top Product
Profit (2015–2017)	Technology	Phones, Copiers	Canon ImageCLASS 2200 Copier
Quantity & Orders	Office Supplies	Binders	Easy-Staple Paper
Sales	Technology/Furniture	Phones, Chairs	Cisco Telepresence, Fellowes Machines

Regional & State Performance

- **Top Region:** West (most years)
- **Top States:** California, New York
- **Lowest States:** Texas, Ohio, Wyoming
- **Weak Regions:** South and Central

Shipping Performance

- **Most Used Mode:** Standard Class
- **Delivery Time:**
 - Standard: 5 days
 - Second Class: 3 days
 - First Class: 2 days
 - Same Day: 0 days

7. Recommendations

1. **Optimize Standard Shipping**
 - Improve logistics to reduce Standard Class delivery time (currently 5 days).
2. **Leverage Seasonal Peaks**
 - Launch marketing campaigns and increase stock before November–December.
3. **Expand High-Profit Categories**
 - Scale up Technology category offerings and marketing (e.g., Copiers, Phones).
4. **Target Underperforming Regions**
 - Improve reach and service in the South and low-performing states via campaigns and infrastructure.
5. **Boost Office Supplies Profitability**
 - Adjust pricing or bundle products to raise profit margins in high-volume categories.
6. **Promote High-Demand Products**
 - Highlight best-sellers like Canon Copiers and Easy-Staple Paper in promotions.
7. **Revive Low-Performing Months**
 - Use January–February for clearance sales and holiday follow-up campaigns.

8. Conclusion

The Superstore Sales Performance Analysis offers a detailed, data-driven view of business performance from 2014 to 2017. It highlights opportunities in product strategy, regional development, shipping optimization, and seasonal marketing. The insights and dashboards created can be used by decision-makers to drive growth and efficiency.

